

Chapter 2 – Software Development Models

Learning Objectives

Understand the purpose of SDLC models, compare major models, identify their advantages and disadvantages, and select an appropriate model based on project requirements.

What are SDLC Models?

An SDLC model defines the framework, sequence of activities, and management approach used to develop software. Different projects require different models depending on complexity, risk, customer involvement, and delivery timelines.

Model	Description	Best For	Advantages	Limitations
Waterfall	Sequential phases completed one after another.	Stable requirements, regulated industries.	Simple, well-documented.	Difficult to accommodate change.
V-Model	Testing activities mapped to each development phase.	Safety-critical systems.	Strong quality assurance.	Rigid and expensive to change.
Iterative	Product is refined through repeated cycles.	Requirements evolve.	Continuous improvement.	Requires active customer feedback.
Incremental	Delivered in multiple functional releases.	Large business applications.	Early value delivery.	Integration complexity.
Spiral	Risk-driven iterations with prototyping.	High-risk, high-cost projects.	Excellent risk management.	High cost and management overhead.
RAD	Rapid prototyping and user feedback.	Small to medium projects.	Fast delivery.	Needs skilled team and continuous user involvement.
Agile	Short iterative sprints with customer collaboration.	Changing requirements.	Flexible, fast feedback.	Requires engaged stakeholders and discipline.

Model Selection Guidelines

Choose Waterfall when requirements are fixed and compliance is important. Choose V-Model when verification and validation are critical. Choose Incremental or Iterative when early releases are valuable. Choose Spiral for high-risk or innovative projects. Choose RAD when rapid prototyping is required. Choose Agile when requirements change frequently and customer collaboration is high.

Real-World Examples

Banking Core System: Waterfall or V-Model due to regulatory requirements. E-commerce Platform: Agile with Incremental delivery. Healthcare System: V-Model for strong validation. Startup Mobile App: Agile or RAD for rapid feature delivery.

Interview Questions

1. Compare Waterfall and Agile. 2. Why is the Spiral model considered risk-driven? 3. When would you choose the V-Model? 4. Differentiate Iterative and Incremental models. 5. What factors influence SDLC model selection?

Chapter Summary

Each SDLC model has strengths and trade-offs. Selecting the right model depends on project size, risk, regulatory needs, customer involvement, team maturity, budget, and expected rate of change.